

Advanced (Habanero)

- Q1.** What is the primary purpose of a stem-and-leaf plot?
- (A) To display the mean of a dataset
 - (B) To show the distribution of a dataset
 - (C) To calculate the median of a dataset
 - (D) To create a box plot
 - (E) To graph a function
- Q2.** Which of the following is a benefit of using a stem-and-leaf plot?
- (A) It can only be used with small datasets
 - (B) It is difficult to read and interpret
 - (C) It can help identify outliers in a dataset
 - (D) It is only used for categorical data
 - (E) It is not useful for comparing multiple datasets
- Q3.** What is the stem in a stem-and-leaf plot?
- (A) The largest value in the dataset
 - (B) The smallest value in the dataset
 - (C) The first digit of each value in the dataset
 - (D) The last digit of each value in the dataset
 - (E) The average of the dataset
- Q4.** Which of the following datasets would be most suitable for a stem-and-leaf plot?
- (A) A dataset with only 5 values
 - (B) A dataset with 1000 values, all between 10 and 20
 - (C) A dataset with categorical values
 - (D) A dataset with only negative values
 - (E) A dataset with all values equal to 0
- Q5.** How do you determine the stem for the value 45?
- (A) The stem is 4 and the leaf is 5
 - (B) The stem is 45 and the leaf is 0
 - (C) The stem is 5 and the leaf is 4
 - (D) The stem is 4 and the leaf is 0
 - (E) The stem is 0 and the leaf is 45
- Q6.** What can be inferred about a dataset if its stem-and-leaf plot is symmetric?
- (A) The dataset is skewed to the left
 - (B) The dataset is skewed to the right
 - (C) The dataset is normally distributed
 - (D) The dataset has a large range
 - (E) The dataset has a small range
- Q7.** Which of the following is NOT a common use of a stem-and-leaf plot?
- (A) To display the distribution of a dataset
 - (B) To compare the distributions of multiple datasets
 - (C) To calculate the mean of a dataset
 - (D) To identify outliers in a dataset
 - (E) To visualize the correlation between two datasets
- Q8.** What is the primary difference between a stem-and-leaf plot and a histogram?
- (A) A stem-and-leaf plot is used for categorical data, while a histogram is used for numerical data
 - (B) A stem-and-leaf plot is used for small datasets, while a histogram is used for large datasets
 - (C) A stem-and-leaf plot shows the actual values in the dataset, while a histogram shows the frequency of each value
 - (D) A stem-and-leaf plot is used for skewed datasets, while a histogram is used for normal datasets
 - (E) A stem-and-leaf plot is used for comparative analysis, while a histogram is used for descriptive analysis

Open Ended Questions (FRQ)

- Q9.** Construct a stem-and-leaf plot for the dataset: 12, 15, 18, 20, 22, 25, 30, 32, 35, 40
- Q10.** Explain how to interpret the distribution of a dataset based on its stem-and-leaf plot. Provide an example.

Chef's Tips

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- Tip 1: Emphasize the importance of accurately constructing the stem-and-leaf plot.
- Tip 2: Encourage students to practice interpreting different types of distributions.
- Tip 3: Remind students to carefully read and follow the instructions for each question.